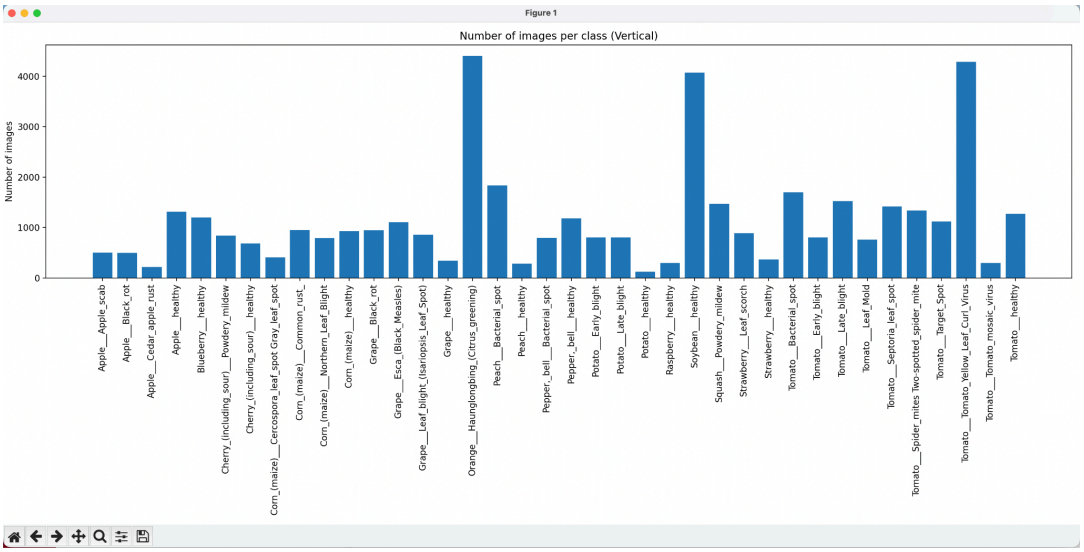
1. Dataset Summary

- PlantVillage dataset: 01_eda.py: **number of classes - 38, total images - 43444**

```
tebooks/01_eda.py
Total classes: 38
Total images: 43444
Apple___Apple_scab: 504 images
Apple___Black_rot: 496 images
Apple___Cedar_apple_rust: 220 images
Apple___healthy: 1316 images
Blueberry___healthy: 1202 images
Cherry_(including_sour)___Powdery_mildew: 842 images
Cherry_(including_sour)___healthy: 684 images
Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot: 410 images
Corn_(maize)___Common_rust_: 953 images
Corn_(maize)___Northern_Leaf_Blight: 788 images
Corn_(maize)___healthy: 929 images
Grape___Black_rot: 944 images
Grape___Esca_(Black_Measles): 1107 images
Grape___Leaf_blight_(Isariopsis_Leaf_Spot): 861 images
Grape___healthy: 339 images
Orange___Haunglongbing_(Citrus_greening): 4405 images
Peach___Bacterial_spot: 1838 images
Peach___healthy: 288 images
Pepper,_bell___Bacterial_spot: 797 images
Pepper,_bell___healthy: 1183 images
Potato___Early_blight: 800 images
Potato___Late_blight: 800 images
Potato___healthy: 121 images
Raspberry___healthy: 297 images
Soybean___healthy: 4072 images
Squash___Powdery_mildew: 1468 images
Strawberry___Leaf_scorch: 887 images
Strawberry___healthy: 364 images
Tomato___Bacterial_spot: 1702 images
Tomato___Early_blight: 800 images
Tomato___Late_blight: 1527 images
Tomato___Leaf_Mold: 761 images
Tomato___Septoria_leaf_spot: 1417 images
Tomato___Spider_mites Two-spotted_spider_mite: 1341 images
Tomato___Target_Spot: 1123 images
Tomato___Tomato_Yellow_Leaf_Curl_Virus: 4286 images
Tomato___Tomato_mosaic_virus: 299 images
Tomato___healthy: 1273 images
```



- **Training set:** 100% of PlantVillage dataset images.
- **Validation set:** 10% of PlantVillage dataset images (randomly selected).
- **Test set:** 10% of images collected from external sources (camera roll, internet) to evaluate generalization.
- **Data cleaning:** Removed blurred or corrupted images from the dataset.

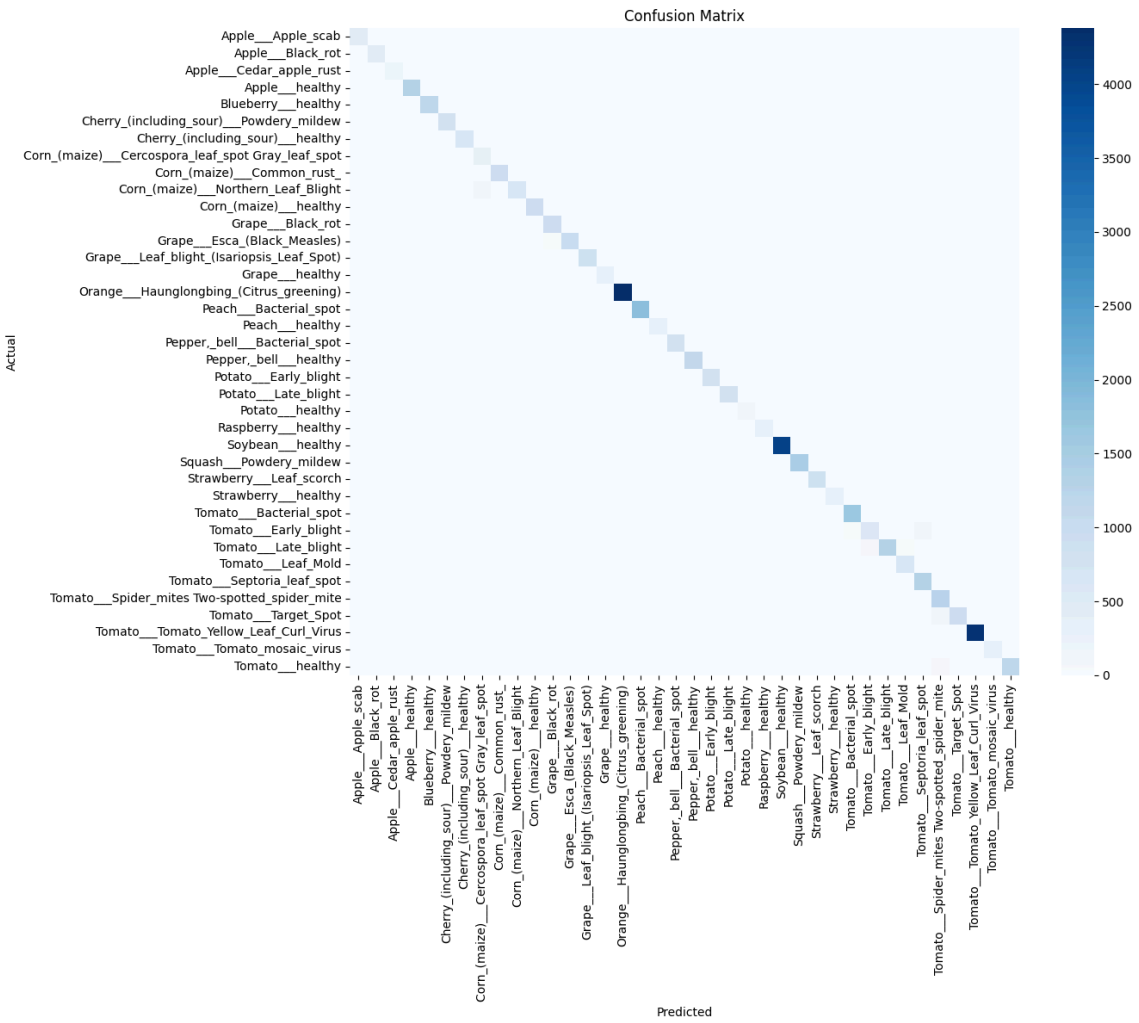
2. Baseline Training: Trained a simple CNN using `train_model.py` with MobileNetV2 (classification head only) for speed, logged accuracy and loss curves, and saved all outputs/plots to the `results/` folder.

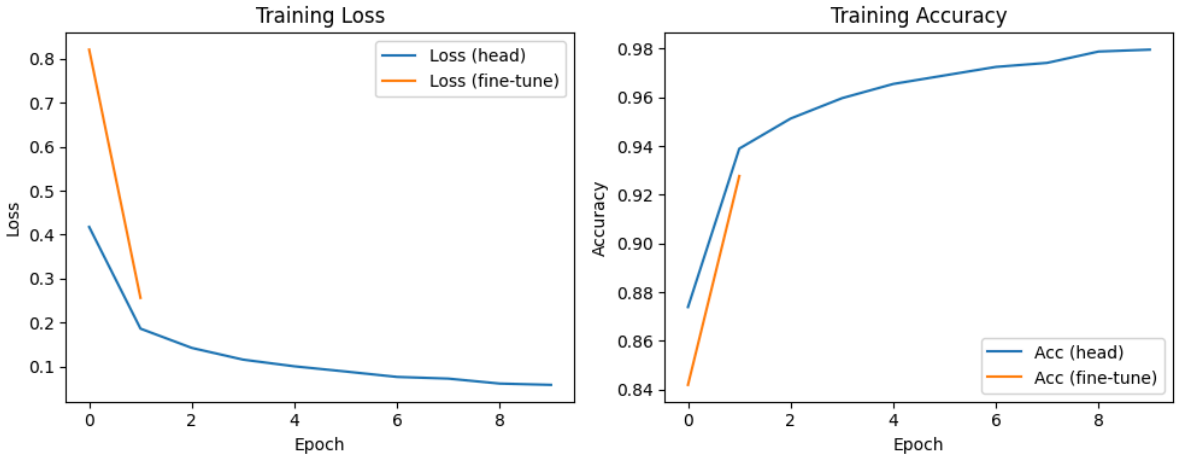
- training_log.csv

logs > training_log.csv > data

```csv
epoch,accuracy,loss,val_accuracy,val_loss

0,0.8748503923416138,0.4144000709056854,NA,NA

1,0.9369993805885315,0.18781012296676636,NA,NA

2,0.9522603750228882,0.14028409123420715,NA,NA

3,0.9605008959770203,0.11657126992940903,NA,NA

4,0.965587854385376,0.10043924301862717,NA,NA

5,0.9706979393959045,0.08544736355543137,NA,NA

6,0.9725623726844788,0.07716088742017746,NA,NA

7,0.9751864671707153,0.07025803625583649,NA,NA

8,0.9802274107933044,0.057361066341400146,NA,NA

9,0.9811021089553833,0.05479481443762779,NA,NA

0,0.8475278615951538,0.7285104393959045,NA,NA

1,0.9296335577964783,0.23811282217502594,NA,NA
```

Training Results





eval.py

```
Macro F1-score: 0.9529

Confusion Matrix:
[[ 465    2    0 ...    2    0    0]
 [   3  475    0 ...    0    0    1]
 [   0    1  214 ...    0    0    0]
 ...
 [   0    0    0 ... 4263    6    0]
 [   0    0    0 ...    4  291    0]
 [   0    0    0 ...    0    1 1168]]

Classification Report:
                                                      precision    recall  f1-score   sup
port

                                Apple___Apple_scab       0.99      0.92      0.95
 504
                                 Apple___Black_rot       0.99      0.96      0.97
 496
                          Apple___Cedar_apple_rust       0.98      0.97      0.98
 220
                                   Apple___healthy       0.98      0.98      0.98
1316
                               Blueberry___healthy       0.97      1.00      0.98
1202
           Cherry_(including_sour)___Powdery_mildew       0.99      0.98      0.99
 842
                  Cherry_(including_sour)___healthy       0.99      0.98      0.98
 684
Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot       0.82      0.92      0.87
 410
                        Corn_(maize)___Common_rust_       0.99      0.99      0.99
 953
                Corn_(maize)___Northern_Leaf_Blight       0.94      0.89      0.92
 788
                             Corn_(maize)___healthy       0.99      1.00      1.00
 929
                                  Grape___Black_rot       0.95      0.99      0.97
```

app.py web based output:

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Predict

Prediction: Pepper,_bell___Bacterial_spot (57.85%)

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Predict

Prediction: Strawberry___healthy (89.11%)

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Choose file No file chosen

Predict

Prediction: Strawberry___healthy (100.00%)

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